ALBERT AROKIYASOURIDASS

Healthcare Solutions Engineer

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PROFESSIONAL SUMMARY

Healthcare Solutions Engineer with 17+ years of experience delivering secure, HIPAA-compliant software. Spent more than a decade embedded with a leading medical records provider, architecting platforms for manual provider retrieval, patient intake, facility discovery, and payer, life sciences, audit, and FHIR workflows—from discovery through production launch—using Azure, AWS, REST APIs, and HIPAA/PHI-governed systems. Since 2024, independently researching generative AI for document intelligence by designing and evaluating RAG pipelines with hybrid BM25 and vector retrieval, LangGraph multi-agent workflows, and IR metrics such as MRR and RRF to improve grounding and reduce hallucinations. Seeking an AI Customer Engineer role that combines proven customer-embedded delivery with applied AI research to lead pilots, partner with healthcare customers, and move solutions from proof of concept to trusted production.

Tech: C#, Python (Django), Node.js, .NET, Azure, AWS, OpenAI, Claude, RAG, Hybrid Retrieval (BM25 + Vector), React, REST APIs

CORE STRENGTHS

- Architected medical record retrieval platforms for a leading provider, including manual retrieval, patient intake, and facility discovery workflows
- HIPAA/PHI governance, secure integrations, and production-launch readiness
- Healthcare data exchange, medical record workflows, clinical audits, and claims reviews
- Solution architecture, REST APIs, and cloud-native systems on Azure and AWS
- Document retrieval using RAG, KAG, CAG, hybrid BM25/vector search, MRR, and RRF
- Pilot-to-production delivery focused on validation, auditability, and go-live partnership with workflow owners
- Translating needs across customers, product, architecture, and engineering teams

PROFESSIONAL EXPERIENCE

Relevantz Technology Services | United States since January 2016 · Alpharetta, GA / Remote

Solution Engineer — Healthcare Platforms · 2012 – Present

Primary client: A leading medical records provider. The engagement centers on record retrieval for payer, life sciences, and audit use cases, including manual provider retrieval, patient intake, facility discovery, FHIR exchange, and HIPAA compliance. End-customer names are confidential.

- Embedded with the client to architect and deliver medical record workflows, from manual provider retrieval to automated, production-ready exchange, translating operational needs into secure technical solutions.
- Partnered across design, implementation, validation, and release cycles, owning technical follow-through in HIPAA/PHI environments from requirements workshops through production readiness.
- Architected and built secure platforms for medical record retrieval and request lifecycle management across distributed provider networks and nationwide healthcare workflows.
- Led solution design for clinical audit and claims review platforms supporting RAC/MAC operations across 4,000+ hospitals, improving compliance traceability and review efficiency.
- Delivered cloud-native healthcare systems on Azure and AWS, including web applications, REST APIs, secure document exchange, and batch integrations with encryption, RBAC, audit logging, and PHI governance.
- Built REST APIs and integration services for exchanging data among hospitals, provider organizations, and enterprise systems under regulated security controls.

• Guided engineering teams through architecture reviews, design, implementation, testing, and production deployment; improved reliability with monitoring, structured logging, and CI/CD on Azure DevOps and AWS.

Selected initiatives

Medical Record Retrieval Platform (OnDemand)

• Designed enterprise workflows to securely manage the medical record request lifecycle across provider networks and healthcare organizations.

• Built scalable REST APIs and integration services that support request submission through portals, APIs, and file-based interfaces.

• Delivered AWS infrastructure for secure document exchange and high-volume healthcare transactions, with QA workflows, audit trails, and compliant delivery.

Clinical Audit & Claims Review Platform (Audapro)

• Implemented secure document intake and claims review workflows on Microsoft Azure for hospital and insurance audit operations.

• Streamlined multi-organization integrations and compliance reporting through structured audit logs and operational dashboards.

OKS Prepress Services Pvt Ltd | Chennai, India

Software Engineer (started as Trainee Software Engineer) · 2009 – 2012

• Designed, developed, tested, and debugged solutions using C#, ASP.NET, ADO.NET, and SQL, advancing from trainee to software engineer over three years.

• Collaborated on troubleshooting, code reviews, and Microsoft-platform best practices, building the .NET foundation later applied across 14+ years of healthcare platform engineering.

AI & INDEPENDENT WORK

AI Document Intelligence Platform · Knowledge Studio · Independent (solo) · 2024 – Present

• Designed and built a document intelligence platform that ingests content from connected sources, chunks and indexes it asynchronously, and lets users query their knowledge while ingestion continues in the background.

• Built a hybrid retrieval pipeline that combines OpenAI embedding-based semantic search with BM25 lexical search in OpenSearch, then fuses independently ranked results using Reciprocal Rank Fusion (RRF) to broaden retrieval coverage.

• Implemented evidence-oriented retrieval that displays the source chunks supporting each answer, allowing users to validate retrieved information instead of relying solely on generated responses.

• Evaluated retrieval quality with information-retrieval metrics, including Mean Reciprocal Rank (MRR), and used ranking analysis to strengthen grounding and reduce unsupported answers.

• Separated document ingestion, chunking, indexing, retrieval, ranking, and conversational access into distinct stages, enabling asynchronous processing and uninterrupted user interaction during ingestion.

TECHNICAL SKILLS

Healthcare domain: Medical record retrieval, manual provider retrieval, patient intake, facility finding, payer, life sciences, clinical audit, FHIR, HIPAA, PHI security

Languages: C#, Python, TypeScript, JavaScript

Backend & APIs: .NET / ASP.NET Core, Python (Django), Node.js, REST APIs, Entity Framework Core, distributed systems

Frontend: React, Angular

Cloud & DevOps: Microsoft Azure, AWS, Azure DevOps, CI/CD, monitoring & structured logging

AI systems & tooling: OpenAI, Claude, Lang Graph, RAG, KAG, CAG, hybrid/vector/BM25 retrieval, MRR, RRF, embeddings, intelligent document processing, grounding evaluation

EDUCATION

Bachelor of Engineering (Electronics & Communication Engineering) — Anna University